EXPERIMENT ID: USIT-V-2026-X1

Title: Experimental Verification of Rotational Pressure Gradients in High-Density Media

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1. Executive Summary

This protocol details the experimental verification of the USIT framework. The experiment is designed to determine if rotational work induces a gravitational pressure gradient (Pg) defined by Pg = (omega^2) * Omega_c. By utilizing a high-density, non-magnetic Steel rotor and vacuum-isolated instrumentation, this protocol establishes a rigorous "Go/No-Go" verification point for the framework.

2. Experimental Parameters (Optimized for Steel)

- Rotor Material: High-density, non-magnetic Steel
- Rotor Mass: 32.88 kg (10cm radius sphere)
- Test Mass (Probe): 1.0 kg
- Distance (r): 0.6096 meters (2 feet)
- Max Rotational Speed: 50,000 rad/s
- Coupling Constant (Omega_c): 1.0e-20 (Architect's adjusted sensitivity)

3. Required Materials & Equipment

- Vacuum Chamber: Required to eliminate air drag and thermal convection.
- Precision Sensor/Torsion Balance: Must be capable of detecting forces in the 10^-12 N range.
- 32.88kg Steel Sphere: Must be precision-balanced to prevent vibration.
- High-Speed Motor System: Industrial-grade controller capable of 50,000 rad/s.
- Isolation Table: Active seismic/vibrational isolation platform.

4. Estimated Budget (USD)

- Vacuum System: $15,000 – $30,000
- Precision Sensor/Torsion Balance: $10,000 – $20,000
- 32.88kg Steel Rotor: $1,000 – $3,000

- High-Speed Motor & Controller: $5,000 – $10,000
- Isolation, Fixturing, & Misc: $5,000
- TOTAL ESTIMATED COST: $36,000 – $68,000

5. Mathematical Expectations (Rotational Force Divergence - RFD)

As the Steel rotor spins, the total force measured at the probe increases according to:

Total Force = Newtonian Force (5.9051e-9 N) + (omega^2 * 1e-20)

- At 0 rad/s: 5.9051e-9 N
- At 25,000 rad/s: 5.9113e-9 N
- At 50,000 rad/s: 5.9301e-9 N

6. Step-by-Step Procedure

1. Calibration: Ensure the noise floor is significantly below 10^-12 N.
2. Setup: Secure the Steel rotor in the vacuum chamber, 2 feet from the probe.
3. Null Test (Test B): Engage systems without rotation to record baseline data.
4. Experimental Test (Test A): Gradually ramp the rotor to 50,000 rad/s.
5. Data Analysis: Compare Test A against Test B. A quadratic upward curve confirms the Rotational Force Divergence (RFD).

7. Expected Results & Conclusion

Expected Results: We expect to observe a "Quadratic Divergence" in the force measured at the probe. As the rotor accelerates, the force will deviate from the constant Newtonian baseline and follow a curve proportional to the square of the angular velocity (omega^2). This pattern constitutes the Rotational Force Divergence (RFD).

Conclusion: This experiment is designed to be a definitive test of the USIT framework. By successfully isolating the rotational nudge from background gravity, we can verify the existence of the predicted pressure gradient. Whether the results confirm the predicted Omega_c value or set a new empirical upper bound, this test provides the necessary data to move the USIT framework from a theoretical model to a verified physical phenomenon.

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